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A LSTM-GAN Algorithm for Synthetic Data Generation of Time Series Data for Condition Monitoring

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Abstract

Condition monitoring plays a crucial role in real-time evaluation of system states, but requires a large amount of measurement data to develop an accurate model. In reality a sufficient amount of data is often not available. Therefore, the proposed approach focuses on data augmentation using a Wasserstein Generative Adversarial Network (WGAN) to augment time series condition data. To enable the generation of synthetic data in the time domain, a Long Short-Term Memory (LSTM) network architecture is used in conjunction with a WGAN. The practical implementation of WGAN with a LSTM architecture is verified with two different datasets, a vibration and an acoustic dataset. Therefore, vibration data of rolling bearings in various system states recorded on a rolling bearing test rig and acoustic data of a welding process are used for synthetic data generation in separately trained networks. This work also focuses on a detailed evaluation of synthetic data in relation to the real data using various methods, such as distribution function analysis, a parameter analysis, and a visual comparison in the time as well as frequency domain. A classification model is also used to classify the real and a combined (real and synthetic) dataset to verify the benefits. The classification of the acoustic data shows an improvement in test accuracy with the combined dataset to 100% compared to the existing real measurement data of 65%. The following paper highlights the potential of the mentioned algorithm for data augmentation as an optimal solution for the described use cases. Synthetic data generation in the time domain is also critically discussed and the difficulties involved are emphasized.

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1. Introduction

Condition monitoring of machines and production processes plays an important role in ensuring high availability, reliability and efficient use of resources. In this context, regular maintenance and servicing are essential for a reliable operation. However, many industrial and automotive systems are not maintained in a condition-based or predictive manner. This leads to unplanned and unpredictable system failures and quality issues. For this reason, condition-based and predictive maintenance strategies are increasingly being used to counteract breakdowns. The implementation and use of machine learning algorithms in relation to condition monitoring means an improvement in the classification of bad and deteriorating system conditions in many applications. Different condition variables, such as vibration signals, can be used to monitor transmissions and thus enable conclusions to be drawn about the condition of a system. An analysis of the measured condition variables is generally possible in the time, frequency or time-frequency domain. An important point for the analysis and the implementation and of condition monitoring systems is the acquisition of condition data, for example with suitable sensors. [1–3]

In many applications, condition data is not available in a sufficient manner. However, the acquisition of condition data is an important issue for the implementation of condition monitoring systems and the training of machine learning algorithms [4]. In reality, the acquisition of data is a particular challenge, as machine learning algorithms cannot be trained without a sufficient amount of data. Generative Adversarial Networks (GAN) allow synthetic data generation, based on real data to extend or augment existing datasets. There is a large number of different GAN algorithms to generate data depending on the respective application [5,6]. Wasserstein Generative Adversarial Networks with Gradient Penalty (WGAN-GP) therefore represent a suitable approach for data augmentation, for example of vibration or acoustic data. Based on real data, relevant features in the measured signals and the real dataset distribution are learned and replicated in the synthetically generated data. The newly generated data are no copies of the real data, but new data with similar properties. In most cases, appropriate pre-processing methods are used to emphasize relevant features and to reproduce or generate signal characteristics as realistically as possible. [7–9]

However, this paper presents an approach to generate synthetic data from time series condition data using a combination of GAN and a Long Short-Term Memory (LSTM) network architecture. A LSTM network is a type of Recurrent Neural Network (RNN) and has the key advantage that it can learn dependencies between time steps of sequence data. Therefore, the research hypothesis is to show the possibility of synthetic data generation directly in the time domain without the extensive use of pre-processing methods and a deep data understanding. In this paper, the mentioned approach is visualized in detail as well as validated with two different datasets. One dataset is characterized by vibration data measured on rolling bearings with different system states. Furthermore, a dataset with acoustic data of a welding process is used. The synthetic data is evaluated in regard to the real data in various ways, such as by comparing the probability distribution as well as with a parameter analysis.

2. State of the Art

The proposed approach focuses on data augmentation using a WGAN in combination with a LSTM network architecture for time series condition data. Condition monitoring systems are used to prevent machine and system failures. Nevertheless, a lot of systems, machines, and processes are not maintained in a condition-based or predictive manner. These applications are maintained with reactive or time-based maintenance strategies. In the future, however, more system and process monitoring will be used, that unforeseen failures can be avoided and a detailed understanding of the system can be achieved. The implementation of monitoring systems requires sufficient measurement data in form of condition variables, which are measured in the most cases via different sensors. For example, the condition of mechanical systems can be monitored by a particle analysis, thermography as well as with an acoustic or a vibration analysis [10]. A combination of these methods is also possible. However, vibration and acoustic analysis has proven to be the most suitable methods for monitoring machines and processes due to the long warning time. [1–3]

Sensor positioning is essential for the acquisition of measurement data for both acoustic and vibration sensors. The sensor position and the excitation from external vibration sources have a significant influence on the quality of the measurement data [11]. Therefore, the position of the accelerometers and the microphones should be as close as possible to the vibration source [12].

In most applications, measurement data from processes or machines is saved, pre-processed and analyzed. The use of pre-processing methods generally depends on the respective application to improve the quality of data or to highlight certain features [13,14]. Data analysis is generally possible in the time, frequency and time-frequency domain. A simple way for evaluating vibration data is to calculate condition indicators, which can be used for parameter analysis, for example. In the time domain, detailed fault detection is difficult, but has the advantage that no detailed understanding of the data and no extensive pre-processing is required. In the frequency domain, however, significant features can be highlighted, for example by an envelope demodulation, to allow conclusions about occurring failures. Another specified form is the time-frequency domain analysis. [3]

The use of machine learning models, such as neural networks, for condition monitoring brings many advantages in the classification of different or deteriorating system conditions [10]. Machine learning based approaches require huge amounts of data. In practice, however, it is difficult to measure the required amount of data for a number of reasons, including cost, labor, and complex sensor technology. Therefore, a sufficient amount of data is often not ensured in reality [15]. GANs have gained considerable attention in recent years due to advances in technology, particularly in the augmentation of existing datasets. A GAN offers benefits in data augmentation and image synthesis [16,17]. However, a normal GAN has some limitations such as mode collapse, non-convergence and decreasing gradients during training [18]. An improved GAN, called WGAN-GP, which enforces the Lipschitz constraint on the critic, penalizes the norm of the critic's gradients to prevent mode collapse and stabilize training, serves as a successful implementation approach [19]. The implementation of a Wasserstein distance loss function into the model ensures the smoothness in gradients and prevents a gradient explosion. Arjovsky et al. described this kind of algorithm [20].

In other papers, the described approach is used to generate synthetic data for fault classification and diagnosis [7–9,21]. These types of algorithms are used to augment unbalanced datasets to enable better fault diagnosis, for example, on rotating machinery [22] or on rolling bearings [23].

With the focus on generating synthetic datasets for time series data for condition monitoring, LSTM network architectures offer a great advantage in back propagation. LSTM networks are a type of a RNN that are widely used for time series and stock prediction [16,17]. With their ability to store long-term time series data, they solve complex challenges with historical time series data using additional built-in memory cells [19]. The large time series data is stored using LSTM cells, which contain mechanisms that regulate the flow of information to the system. However, due to conventional training, the possibility of overfitting, extensive computation, large model sizes are their drawbacks [24,25]. On the other hand, their ability to capture temporal dependencies and long-term dependencies in time series data has proven their potential [21]. For example, the existing state of the art focuses on WGAN-GP network architectures with LSTM layers and includes a combination of LSTM layers with convolutional layers [24,25]. Thus, a combination of LSTM layers with WGAN-GP architectures is used for time series anomaly detection for prognostics an health management systems [26] and for melody generation from lyrics [27,28].

For the generation of synthetic data that replicates real data, the integration of advanced GAN algorithms in combination with LSTM network architectures offers a novel approach to address the challenges described above. This synergy not only addresses limitations of LSTM in terms of data complexity and computational efficiency but also enhances the model's ability to generate time series data.

3. Proposed LSTM-WGAN network architecture

The LSTM network is designed to overcome the limitations of a traditional RNN, especially in handling long-term dependencies. LSTM networks, with their fundamental architecture of memory cells, demonstrate enhanced performance in handling transient time series data. These cells are designed to retain information for extended periods of time, significantly reducing the need for pre-processing steps. This characteristic makes LSTM networks particularly useful for applications where the sequential and temporal distribution of data is important.

Figure 1 depicts the LSTM based WGAN-GP architecture where both the generator and the critic consist of LSTM layers and work adversarial to each other. A noise vector is used as an input to the generator and passes the time series data to the LSTM layers one after the other, taking the window length into account. The generator therefore tries to generate synthetic data with a probability distribution similar to the real ones. The critic tries to differentiate between real and synthetic data. Both networks try to optimize their losses. If the generated data is close to the real data and

therefore hard to differentiate, the loss of the generator is reduced and the loss of the critic rises. Following, the generated synthetic data is evaluated in regard to the real data. [19,20]

The algorithms are implemented in python using various libraries such as TensorFlow. The algorithms are trained and evaluated using a personal computer. The basic network architecture is described in several publications [5,6]. The hyperparameters of the specified architecture are defined based on empirical studies with different architectures.

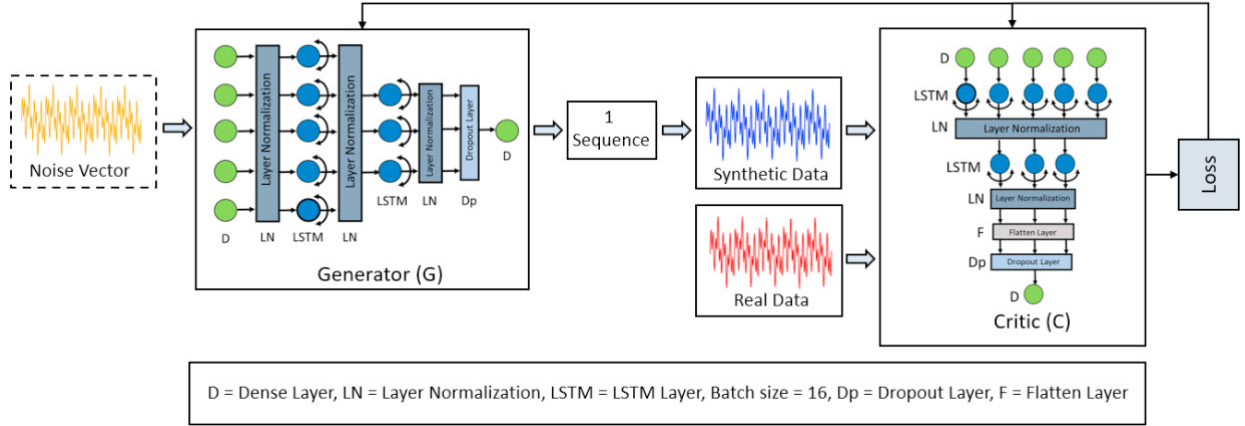


Figure 1: Proposed LSTM WGAN-GP network architecture

The used Wasserstein loss function prevents mode collapse and stabilizes the training. Thus, the Wasserstein distance can help the model to ensure gradient smoothness. The loss function is expressed in equation (1) and produces a continuous output [20]. [19]

$$\min_G \max_D E_{x \sim P_r} [D(x)] - E_{\tilde{x} \sim P_G} [D(\tilde{x})] \quad (1)$$

Here, $\tilde{x} \sim P_G$ is the probability score of synthetic samples and $x \sim P_r$ is the probability score of real samples. The generator G tries to minimize the difference between the critic output value for the real data $D(x)$ and the critic output values for the generated data $D(\tilde{x})$, while the discriminator tries to maximize it [9,19]. The Wasserstein distance $W(P_r, P_G)$ also known as the earth-mover distance, is conceptually described as the least amount of effort required to convert distribution P_r into P_G . $W(P_r, P_G)$ maintains the continuity over all points and exhibits differentiability in most cases. For a stable training it is essential, that the critic follows the Lipschitz constraint with a gradient penalty. Equation 2 describes the function for the critic output. The critic evaluating the generated synthetic data is penalized with the gradient penalty norms. [19]

$$L = E_{\tilde{x} \sim P_G} [D(\tilde{x})] - E_{x \sim P_r} [D(x)] + \lambda * E_{\hat{x} \sim P_{\tilde{x}}} [(||\nabla_{\hat{x}} D(\hat{x})||_2 - 1)^2] \quad (2)$$

4. Experimental Verification

4.1. Vibration dataset of bearings

The vibration dataset referenced in this paper is derived from experiments conducted on a bearing test rig. A roller bearing of type 6006, on which an accelerometer (PCB-603C01) is mounted, is installed on the test rig. The used acceleration sensor has a measuring range of $\pm 490 \text{ m/s}^2$, a sensitivity of $10.2 \text{ mV}/(\text{m/s}^2)$, and a frequency range of 0.5 Hz to 10,000 Hz. Data acquisition is carried out with a constant sampling rate of 51,200 Hz, with constant radial and axial loads and a constant motor speed of 1,000 rpm. A radial force of 1,000 N is applied via a pulley system, as shown in Figure 2 (a).

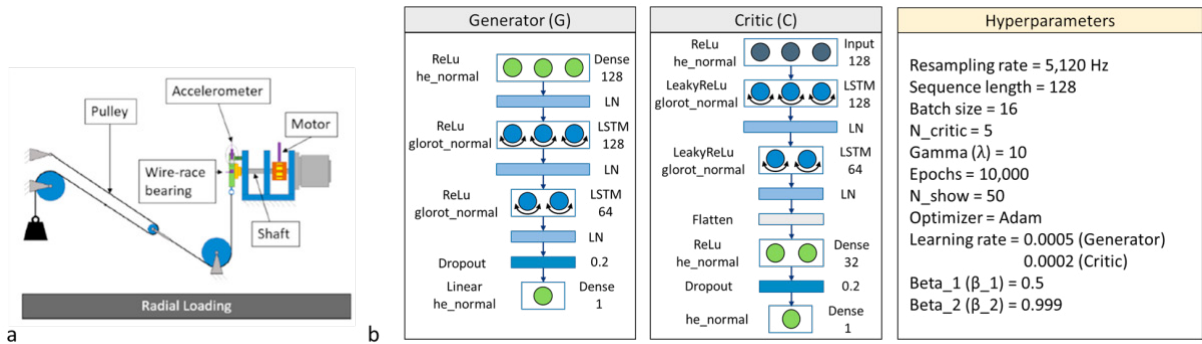


Figure 2: (a) Experimental setup for bearing test rig & (b) Detailed architecture design of LSTM WGAN-GP for acceleration data

A total of 4 bearings are available for the acceleration measurements, each with a different system condition. The system conditions are described in detail in Table 1. A total of 10 measurements of 1 second are available from each bearing. Therefore, a total of 40 measurements were carried out.

Table 1: Labels with respective system states of the bearing

	System condition	Description
Label 1	Healthy	New bearing with a healthy state
Label 2	Faulty	Faulty bearing with realistic failure occurring during operation
Label 3	Faulty	Faulty bearing with discrete outer ring damage by lasering
Label 4	Faulty	Faulty bearing with chemically induced corrosion

Under the existing conditions of the dataset, it is not possible to train a model to classify the different states of bearings due to the insufficient amount of data. Implementing LSTM based WGAN-GP for data augmentation is experimented to enhance the existing datasets.

Figure 2 (b) depicts the detailed architecture of the LSTM WGAN-GP for acceleration data. The network architecture is designed to generate synthetic vibration signals that closely replicate the real-world conditions. The generator, consisting of LSTM and dense layers, has the function of generating synthetic vibration data based on a noise vector with a length of 50. The sequence length is 128 data points for the generator to ensure a detailed representation of the vibration distributions in each sequence. Each sequence of the generator output is normalized and each time stamp of the sequence is generated separately. The critic, using LSTM and dense layers, evaluates the generated synthetic data against real data and guides the generator to produce increasingly realistic signals. A layer normalization (LN) is carried out between some layers of the generator and the critic. This results in a more stable training. The training also provides the average real score, the average fake score and the Wasserstein distance to ensure a stable training. This adversarial training process is optimized with Adam for both the generator and the critic, applying a gradient penalty. The parameters used in the training process, as visualized in Figure 2 (b), are determined empirically. The training process is defined over 10,000 epochs with periodic evaluations every 50 epochs. A batch size of 16 is defined and the critic is trained 5 times for each training of the generator (N_{critic}), emphasizing the role of the critic in the adversarial training process. The implementation of a gradient penalty, controlled by a lambda parameter (λ) set to 10, ensures that the critic's gradients remain stable throughout training, preventing a mode collapse.

The acceleration time history is visualized in Figure 3. A comparison of real and synthetic generated data for 1 second is shown. Label 1 represents the good system state, in which the amplitudes are between -2 m/s^2 and 2 m/s^2 . Label 4 shows a faulty bearing with chemically induced corrosion, with amplitudes between -6 m/s^2 and 6 m/s^2 . It can be seen that the real and synthetic data overlap under both conditions, but a direct comparison of the distribution and characteristics in the time domain is not possible.

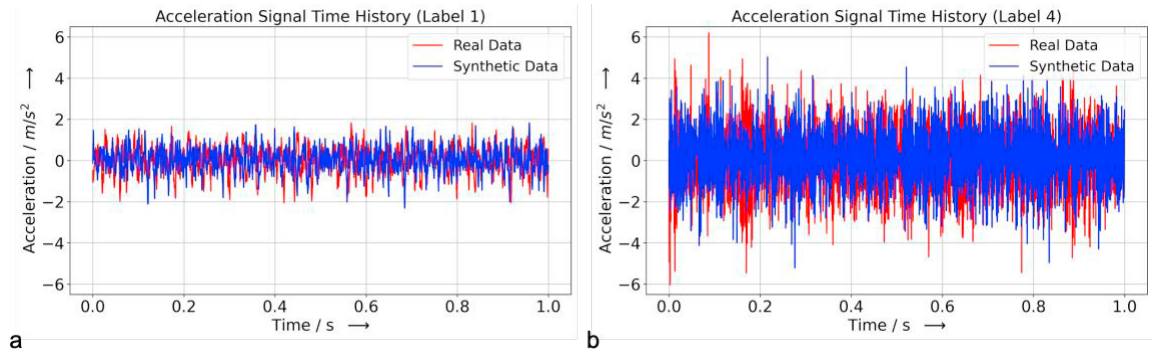


Figure 3: Acceleration signal time history for real and synthetic data (a) Label 1 at 9500 epochs and (b) Label 4 at 9950 epochs

Figure 4 depicts the distribution function of real and synthetic generated data for Label 1 and Label 4. It can be seen that the distribution function of the real data can't be exactly reproduced for both visualized labels. However, a sufficient approximation can be achieved, especially for Label 1.

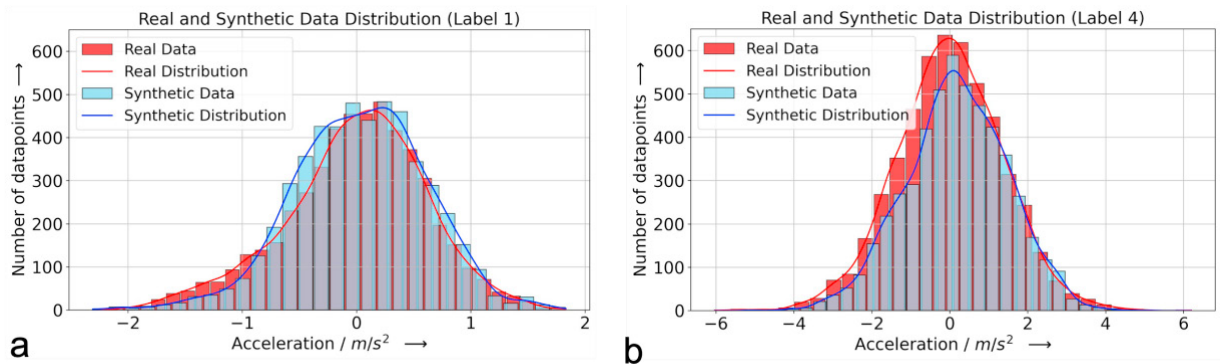


Figure 4: Distribution function for the amplitudes of acceleration for comparison of real and synthetic data (a) Label 1 and (b) Label 4

A parameter analysis, such as analyzing the Root Mean Square (RMS) and the Kurtosis values, is also done. A segment of 128 data points is considered as a window through 1 second of data for all labels as shown in Figure 5.

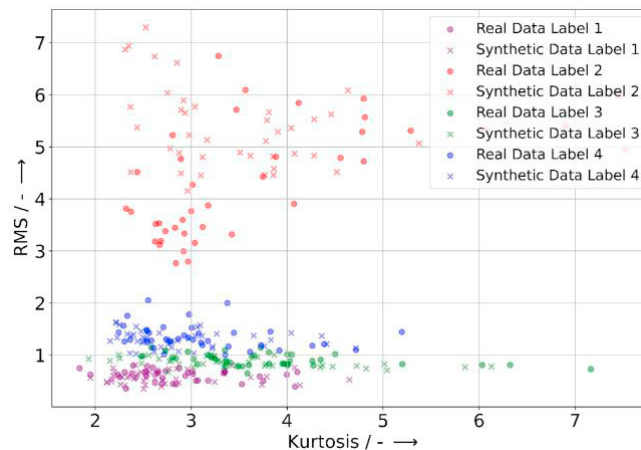


Figure 5: RMS versus Kurtosis analysis for all labels

Figure 5 shows that the synthetically generated data of labels 1, 3 and 4 replicate the relevant features so that the features are learned in comparison to the real data. Although the main features in the data of label 2 could be learned by the network, the synthetically generated data is not converging exactly to the real data.

Furthermore, these generated signals are also analyzed using a time-frequency domain analysis with spectrograms, as shown in Figure 6. It can be observed that the WGAN can learn the characteristic features in the data at higher epochs. The characteristic frequencies and amplitudes in the synthetically generated vibration data are close to those in the real data.

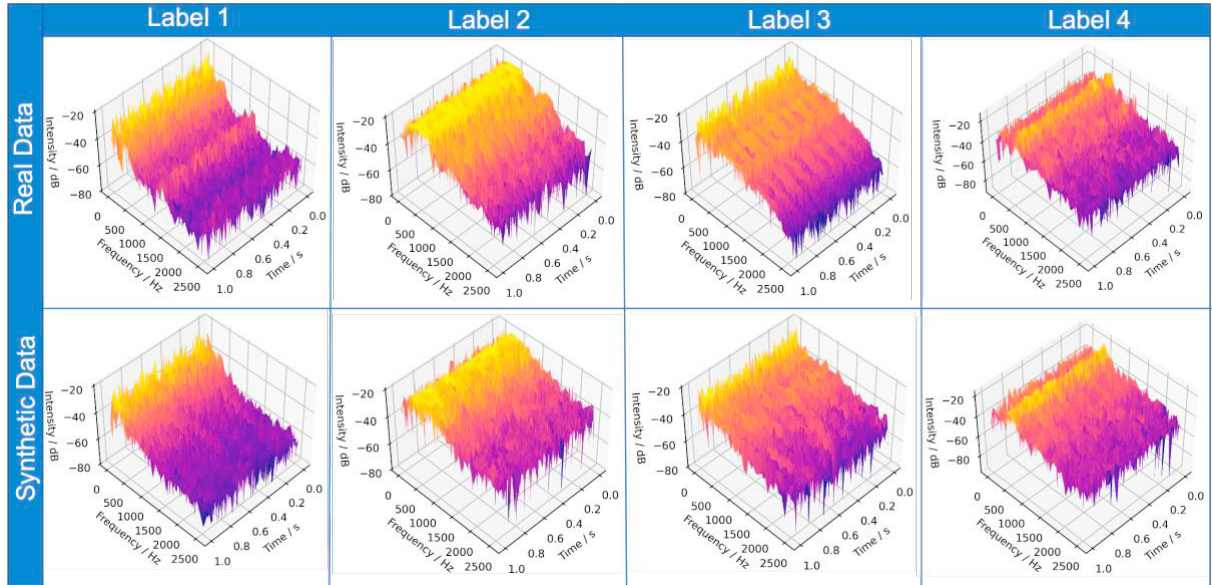


Figure 6: Spectrogram analysis for the different labels (Comparison of real and synthetic data)

4.2. Acoustic dataset of a welding process

Automated strip welding is used in tank manufacturing to weld different steel sheets together. The process involves the use of rollers to drive stainless steel sheets along a circumferential support, where fixed position electrodes weld the sheets together. The distance between the plates varies during the rolling process, and the distance between the electrodes also affects the quality of the weld. To optimize the process, microphones (PCB 130F20) are placed near the electrodes to monitor noise during welding. The microphones with integrated preamplifiers offer a sensitivity of 45 mV/Pa and a dynamic range of 120 dB, covering a frequency range of 20 Hz to 20 kHz. The test setup is depicted in Figure 7 (a) [29]. Two microphones are positioned on either side of the welding area to monitor the welding process: Inside mic 2 and Outside mic 1. Preliminary studies have shown that microphone 1 provides sufficient information for distinguishing between good and bad welds. Therefore, this microphone is used for further analysis.

Measurements are carried out on 42 samples of good welds and 30 samples of bad welds, whereby each measurement has a duration of 1 second. Considering the continuous nature of the process, the high cost and labor intensity of producing poor welds, a data augmentation strategy using LSTM WGAN-GP is used to augment the dataset. For the model training, 20 samples of each class are used to generate synthetic data. Figure 7 (b) depicts the used network architecture in detail. Each network is trained separately to generate synthetic data for the good and bad welds. Using the proposed architecture, the model is trained for 11,000 epochs.

In addition, the synthetically generated data is evaluated in the time domain and the frequency domain using Fast Fourier Transform (FFT). A classification model is also used to visualize the improvement in classifying the good and bad welds with the real and augmented datasets.

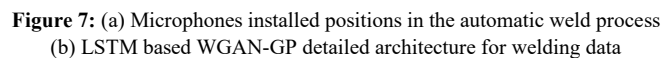


Figure 10 displays four plots comparing Real Data (red line) and Synthetic Data (blue line) for a 'Good Weld' (left column) and a 'Bad Weld' (right column). The top row shows the Time Domain (Amplitude / Pa vs Time / s), and the bottom row shows the FFT (Magnitude / Pa vs Frequency / Hz). Each plot includes a small inset showing the full sequence of data.

Time Domain (Top Row):

- Good Weld (Left):** The Time Domain plot shows a relatively smooth signal with low-frequency components. The inset shows a single sequence for visualization.
- Bad Weld (Right):** The Time Domain plot shows a highly oscillatory signal with significant high-frequency components. The inset shows a single sequence for visualization.

FFT (Bottom Row):

- Good Weld (Left):** The FFT plot shows a dominant peak at approximately 200 Hz, with a smaller peak at approximately 1000 Hz. The Real Data and Synthetic Data are closely aligned.
- Bad Weld (Right):** The FFT plot shows a dominant peak at approximately 200 Hz, with a smaller peak at approximately 1000 Hz. The Real Data and Synthetic Data are closely aligned.

Figure 8: Evaluation of the synthetic data in regard to real data for the welding process: Synthetic data for the good weld is generated at 10,100 epochs and for the bad weld at 3,100 epochs

The original datasets of good and bad welds are divided into training and test data, so that 20 training and 10 test data per class are used for the further investigations. 30 synthetic data for both good and bad welds are generated with LSTM based WGAN-GP network architecture for the classification model based on the 20 real training data. For comparison, the classification model is trained in two ways: On the one hand, only the 20 real training data per class is used, while on the other hand, the augmented dataset with the 20 real data and 30 synthetic data per class is used. Then, 10 training data per class, unseen by both the GAN network and the classification model, are used to validate the classification accuracy. The comparison of the test accuracy of the real and augmented dataset is depicted in Figure 9 using a confusion matrix. Training with the augmented dataset results in a higher accuracy.

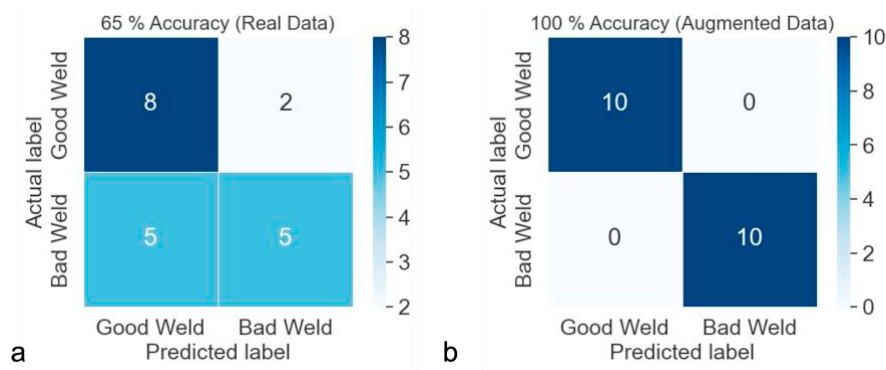


Figure 9: Model test accuracy comparison via confusion matrix (a) Real Data & (b) Augmented Data with LSTM WGAN-GP

5. Discussion and Conclusion

The proposed approach shows an effective way to augment time series data. It can be seen that for both datasets, an acoustic as well as a vibration dataset, synthetic data generation is possible. The relevant features and data distributions can be learned for most of the data in both datasets. However, especially for label 2 of the bearing vibration data set, the synthetic generated data does not exactly converge to the real data, which is probably due to the selected sequence length of the network. Relevant features in the data cannot be learned if the sequence length is too short. In summary, the use of this approach for data augmentation provides a practical solution that improves data quality and facilitates better model training without the need for costly and labor-intensive data acquisition. A more precise classification of system conditions is possible, which enables more reliable quality control in manufacturing processes. However, it can be said that the complex network architecture is a limitation for a higher number of training epochs due to the long training time of such networks, especially if the sequence length is too long. Another limitation is that synthetic data generation in the time domain results in difficulties in highlighting relevant features in relation to the frequency domain. This can lead to poorer data quality.

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